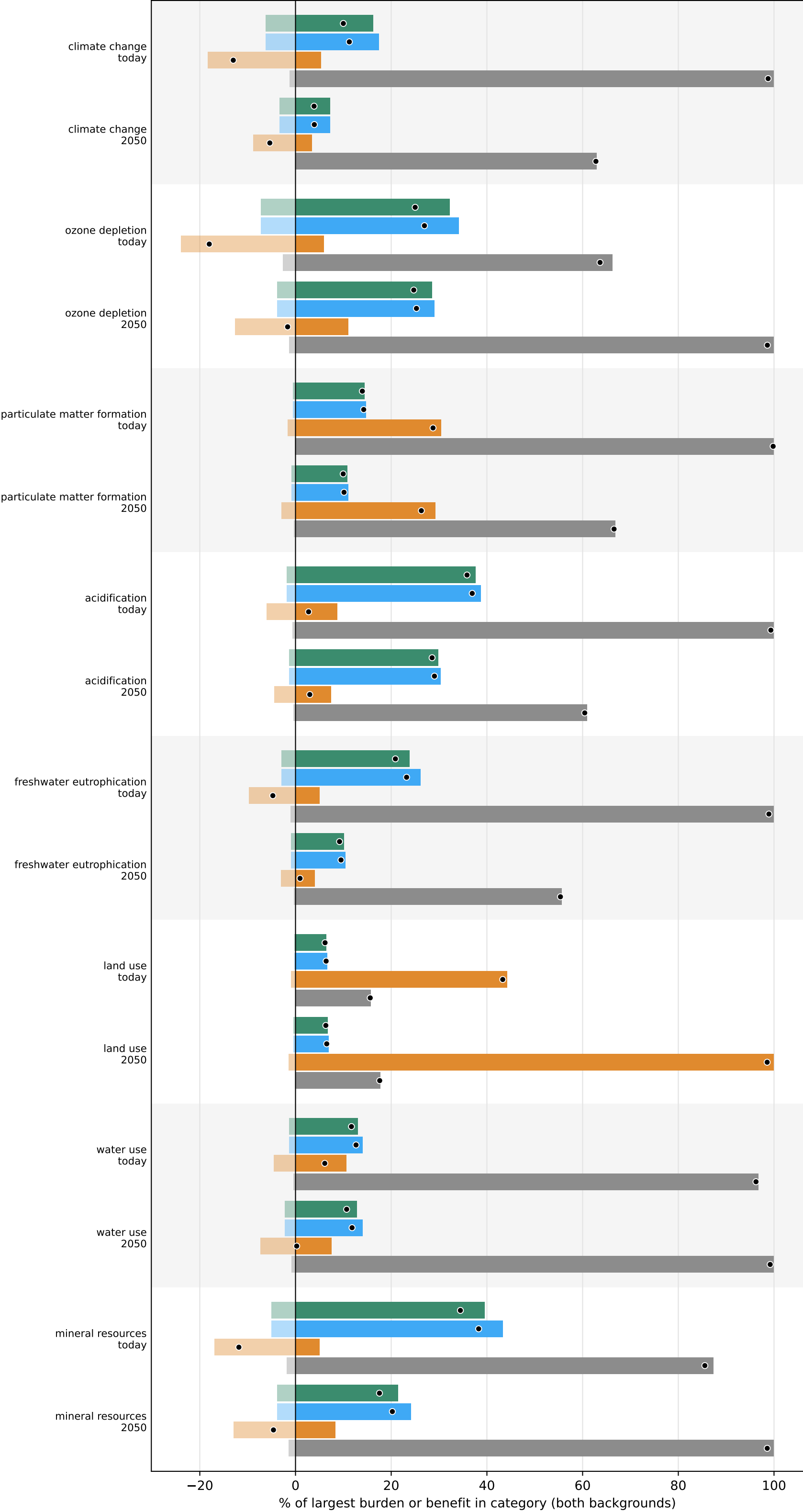


RAW coreless filament winding (beam): today vs. 2050
beam span = 6 m, reference service life 50 yr



RAW coreless filament winding (beam)
RAW coreless filament winding (beam) - scaled up
glulam beam
reinforced concrete beam
benefits (lighter, left)
net impact